



Multiple Protein Subcellular Locations Prediction Based on Deep Convolutional Neural Networks with Self-Attention Mechanism

Hanhan Cong^{1,2} · Hong Liu^{1,2} · Yi Cao^{3,4} · Yuehui Chen^{3,4} · Cheng Liang¹

Received: 28 August 2021 / Revised: 6 December 2021 / Accepted: 13 December 2021 / Published online: 23 January 2022
© International Association of Scientists in the Interdisciplinary Areas 2021

Abstract

As an important research field in bioinformatics, protein subcellular location prediction is critical to reveal the protein functions and provide insightful information for disease diagnosis and drug development. Predicting protein subcellular locations remains a challenging task due to the difficulty of finding representative features and robust classifiers. Many feature fusion methods have been widely applied to tackle the above issues. However, they still suffer from accuracy loss due to feature redundancy. Furthermore, multiple protein subcellular locations prediction is more complicated since it is fundamentally a multi-label classification problem. The traditional binary classifiers or even multi-class classifiers cannot achieve satisfactory results. This paper proposes a novel method for protein subcellular location prediction with both single and multiple sites based on deep convolutional neural networks. Specifically, we first obtain the integrated features by simultaneously considering the pseudo amino acid, amino acid index distribution, and physicochemical property. We then adopt deep convolutional neural networks to extract high-dimensional features from the fused feature, removing the redundant preliminary features and gaining better representations of the raw sequences. Moreover, we use the self-attention mechanism and a customized loss function to ensure that the model is more inclined to positive data. In addition, we use random k -label sets to reduce the number of prediction labels. Meanwhile, we employ a hybrid strategy of over-sampling and under-sampling to tackle the data imbalance problem. We compare our model with three representative classification alternatives. The experiment results show that our model achieves the best performance in terms of accuracy, demonstrating the efficacy of the proposed model.

✉ Hong Liu
lhsdcn@126.com

Hanhan Cong
ado_conghh@ujn.edu.cn

Yi Cao
ise_caoy@ujn.edu.cn

Yuehui Chen
yhchen@ujn.edu.cn

Cheng Liang
alcs417@hnu.edu.cn

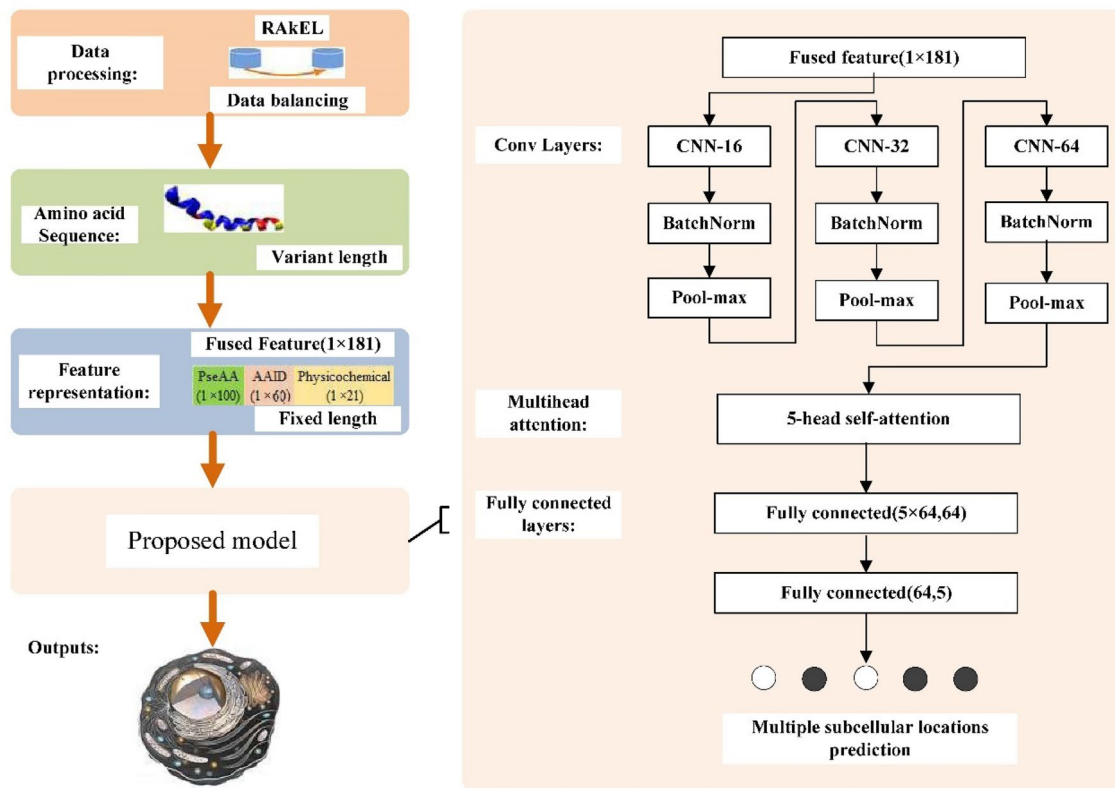
¹ School of Information Science and Engineering, Shandong Normal University, Jinan, China

² Shandong Provincial Key Laboratory for Novel Distributed Computer Software Technology, Jinan, China

³ School of Information Science and Engineering, University of Jinan, Jinan, China

⁴ Shandong Provincial Key Laboratory of Network Based Intelligent, Computing University of Jinan, Jinan, China

Graphical Abstract



Keywords Protein subcellular localization · Deep convolutional neural networks · Attention mechanism · Random k-label sets · Multi-label classification

1 Introduction

Cells are the basic structural and functional units that constitute living organisms. They are the material foundation of life activities [1]. Inside a cell, there are smaller structures that conduct certain functions called subcellular structures [2]. According to their functions and locations, they can be divided into dozens of categories. Different biological species have different subcellular structures [3]. For instance, chloroplast exists in plant cells, but not in human cells. Human cells contain cytoplasm, nucleus, cell membrane [4], etc. As an essential component of cells, proteins participate in many life activities such as DNA transcription and translation, and play critical functional roles in maintaining tissue growth, renewal, and repair [5]. Proteins rely on a specific chemical environment to complete their biochemical reactions so that they can function [6]. Specific subcellular structures provide the chemical environment [7] in which the reaction occurs. Therefore, proteins will be transported to specific subcellular locations under the guidance of the sorting signal after undergoing translational synthesis [8]. If

a protein reaches a wrong subcellular location, it cannot perform normal biological functions [9] and will cause abnormalities in organisms [10]. In essence, medical research has discovered and confirmed a variety of diseases caused by incorrect protein subcellular localization. For example, diabetes is caused by protein deposition and degradation in the endoplasmic reticulum [11]. Therefore, the analysis of protein subcellular locations has significance in biology and medicine. Specifically, by studying the subcellular locations, we can better know the function, evolution, interaction of proteins, and the regulation mechanism in life [12]. Besides, it helps to understand the information of gene function expression [13]. It can also clarify disease pathogenesis [14] and help find a cure [15] or develop new drugs [16].

The protein subcellular location prediction was initially studied through biological experiments [17]. Although traditional experimental methods are of high accuracy and reliability [18], they are time-consuming and cannot scale to large datasets [19]. With the rapid increases of protein sequences, it is impossible to complete protein subcellular localization by biological experiments. Therefore, machine

learning methods are applied to predict protein subcellular locations [20, 21]. The basic idea of machine learning methods is to assign subcellular structures to newly discovered proteins based on the rules learned from the proteins with known subcellular structures [22]. Till now, a number of machine learning methods have been developed to solve this problem. Some of them exploit uncharacterized protein information, while others use the whole proteome sequence information to create new machine learning algorithms for different purposes. Based on the protein sequence information used, existing machine learning methods can be roughly divided into five categories. The fundamental category is called the sequence feature-based methods which only need sequence information. These methods usually explore the correlations of amino acid compositions to identify the protein's biological and functional characteristics [23]. The second is homology-based methods that find highly homologous sequences of the target sequence from the database and make predictions based on the subcellular localization information of the homologous sequences [24]. Therefore, it should be used with caution when applied to nonnative sequences or in the cases that homology is not considerable. Another category of methods makes use of protein domains and motif information. Though not all these motifs and domains are specific to subcellular localization, studying proteins at a domain or a motif level allows more accurate functional inference [25]. The signal peptide-based methods are more restricted since not all proteins contain signal peptides. Thus, they cannot be applied for proteome scale prediction [26]. Other methods leveraged additional data sources such as structural information, fluorescence microscopy images, expression data [27], etc. These methods could be categorized into non-sequence feature-based methods and are designed for specific tasks. Although the sequence feature-based methods only exploit sequences without any additional useful information, they are more practical and cost-efficient. In essence, many machine learning tools have been successfully employed for protein subcellular localization predictions using the sequence information. For instance, there are several protein subcellular localization methods available that used the support vector machine (SVM) as the prediction tool [28, 29]. Random Forest (RF) is an ensemble learning framework that can also effectively analyze and classify complex biological data samples [30]. Recently, deep neural networks (DNNs) are widely applied on subcellular localization prediction tasks due to its powerful representation learning abilities [31, 32].

The protein sequences are represented by strings containing 20 different letters [33], where each letter represents an amino acid residue. To use machine learning models, these raw sequences need to be transformed into vectors or matrices [34]. Specifically, the properties of a protein are closely related to its amino acid residue composition [35]. We need

to extract sequence characteristics by expressing amino acid composition, and then use these characteristics to complete protein subcellular location prediction [36]. Therefore, different feature extraction method reflects different aspects of proteins [37–39]. In protein subcellular locations prediction, different subcellular structures may further correlate with each other [40]. As a result, a single protein sequence feature extraction method cannot fully reflect the characteristics in every category [41]. Besides, a protein sequence was initially considered as being only in a single subcellular structure. Nevertheless, with the gradually deepening of biological research, it has been discovered that proteins might be in multiple subcellular structures [42, 43] and the protein subcellular location prediction is in essence a multi-label classification problem [44]. The classifiers based on binary classification can hardly achieve satisfactory results on multi-label tasks. For instance, Wei et al. used sequence information and Stacked Auto-Encoder (SAE) as the classifier for multiple protein subcellular localization prediction [45]. However, the prediction accuracy of each location differs by several times in their dataset. The same problem existed when using other information to perform the prediction. Cheng et al. adopted GO information and Gaussian Kernel Regression (GKR) as the classifier [46]. Unfortunately, this method could only obtain high accuracies on majority samples. As a result, several difficulties need to be resolved in multiple protein subcellular locations prediction:

1. The associations between protein sequences and multiple subcellular locations are indeed very complicated. More powerful feature extraction models are needed to capture the complex local correlations and long-distance dependencies of amino acids.
2. Most of the traditional machine learning models are binary classification models. They can hardly achieve satisfactory performances in multi-label classification tasks as some challenges, e.g., data re-division. Therefore, it is necessary to build models that can handle one-to-many classification tasks.
3. The protein subcellular locations are highly imbalanced, and more effective methods are required to solve this problem. For instance, in the benchmark dataset, 3360 protein sequences located in cytoplasm and 1054 protein sequences located in secreted, respectively. Only 159 protein sequences are located in both cytoplasm and secreted.

This paper uses feature fusion to comprehensively represent sequences and adopt the deep convolutional neural network (DCNN) [47–49] to minimize the redundant information between individual features by extracting high-dimensional features. In addition, we use Random *k*-label Sets (RAkEL) [50] to retain the number of predicted

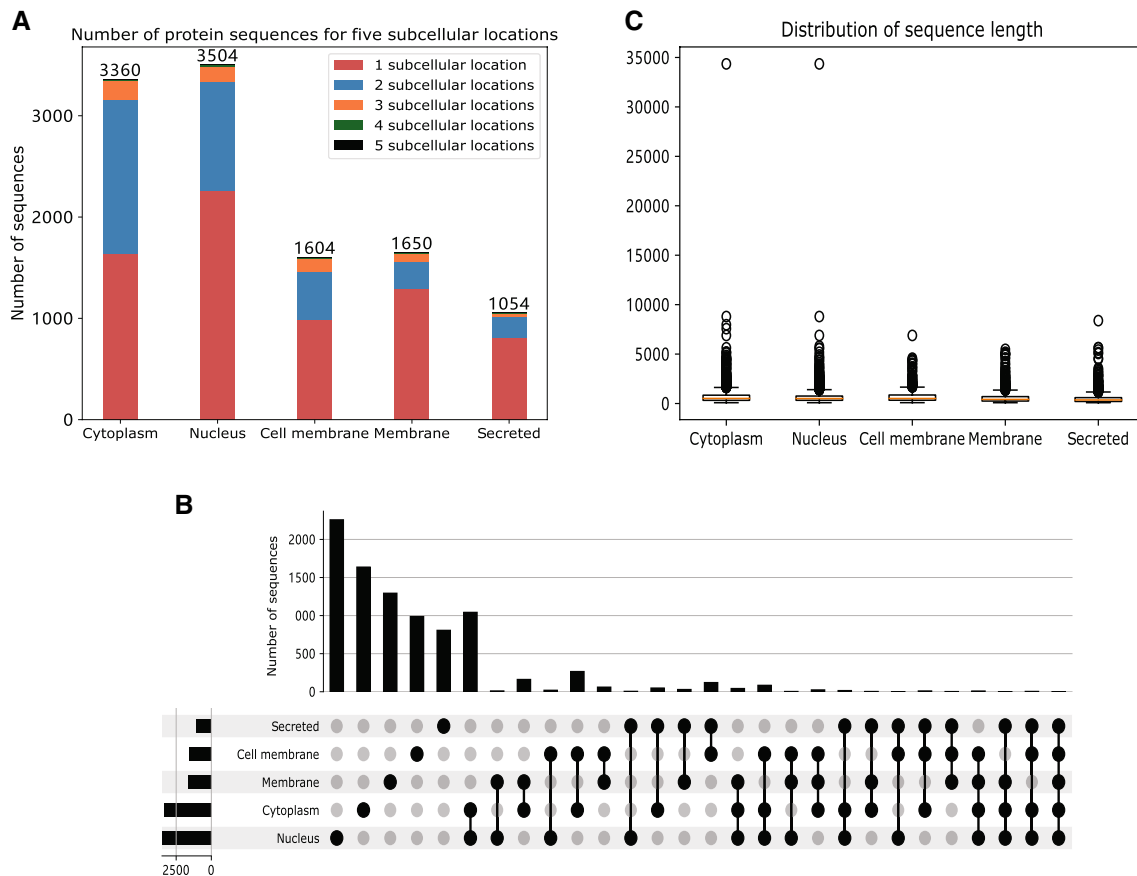


Fig. 1 The statistics of the benchmark dataset

subcellular locations. RAKEL can significantly reduce the complexity of the model training process, but it will also increase the number of negative samples [51]. The classification model tends to produce the label corresponding to the categories with more samples [52, 53]. We employ a combined strategy of over-sampling and under-sampling to generate a balanced training set. Additionally, we use a multi-head self-attention mechanism [54] and a customized loss function. The multi-head self-attention mechanism can further extract the correlation between features. The loss function assigns different weights to positive data and negative data. Through weight adjustment, it ensures that the model is more inclined to positive data. The final prediction model is an ensemble predictor to integrate the merits of different groups, where a best classifier selection mechanism is applied.

2 Materials and Methods

2.1 Dataset

The benchmark dataset is collected from the well-known database UniProtK (<https://www.uniprot.org/>) [55]. The raw dataset is pre-processed by the CD-HIT program [56] with a cutoff value of 0.7 to reduce the sequence similarity. The resulting dataset contains 5 subcellular locations, i.e., cytoplasm, nucleus, cell membrane, membrane, and secreted, with 9053 protein sequences in total. To better capture the sequence information, we filter out the sequences shorter than 81 of amino acid residues. Finally, we get the benchmark dataset with 8963 human protein sequences. The statistics of benchmark dataset is shown in Fig. 1.

Figure 1A shows the number of protein sequences for the five subcellular locations. Each bar represents a location, whose height represents the number of sequences at that location. Specifically, 6984 sequences have single subcellular location, 1979 sequences distribute in two or more subcellular locations simultaneously. We further examine the Multiplicity Degree (MD) of benchmark dataset, which is an estimator measuring the average number of labels per data sample [57] defined by:

$$MD = \frac{N_{loc}}{N} = \frac{\sum_{i=1}^N n(i)}{N} \quad (1)$$

In this definition, N is the total number of protein sequences, N_{loc} is the total number of subcellular locations, and $n(i)$ is the number of subcellular locations marked on the i -th protein sequences. The higher the value of MD , the more proteins with multiple labels. The MD of the benchmark dataset is 1.246.

Figure 1B is an UpSet plot, illustrating the detailed number of protein sequences in each intersection group. The intersection group is presented in the order of the compartment numbers. The upper bar plot presents the number of protein sequences in each intersection group, while the bottom dots show the components of each group. Figure 1C shows the distribution of sequence length for the five subcellular locations.

To compare with state-of-the-art methods, we also use HumB dataset and HumT dataset. The former is divided into two datasets, the training set and the validation set. The latter serves as an independent test set. There is no crossover between the two sets. They are collected from SWISS-PROT with a specific annotation. This paper focuses on 6 subcellular locations, including cytoplasm, extracellular, Golgi apparatus, mitochondrion, nucleus, and plasma membrane. The HumB dataset includes 2838 protein sequences, 2260 of which have single subcellular, and the rest are multi-labeled samples. HumB dataset has a total of 3466 labels and its MD is 1.221. The HumT dataset includes 335 protein sequences, 269 of which have single subcellular, and the rest

are multi-labeled samples. HumT dataset has a total of 408 labels and its MD is 1.218.

2.2 Data Processing

Random k -label Sets We use Random k -label Sets (RAkEL) with overlap strategy [58] to divide the dataset. Given a label set L containing M labels, each label stands for a subcellular localization. There are $\binom{M}{k}$ sets contain k labels ($k \leq M$), each of which is called a k -label set. Let L^K represent the collection of all possible k -label sets. We randomly select N_k sets from L^K so that all the labels appear in these selected sets can cover the label set L . The selected sets form a new collection of k -label sets and is denoted as L^k . Then, for a k -label set L_i^k in L^k , a classifier h_i is constructed. Since one subcellular localization j can be in multiple selected k -label sets, multiple classifiers may produce different predictions for this localization. To resolve the confliction, the prediction made by the classifier with the highest accuracy for that localization j is deemed as the final decision.

Data balancing strategy: according to the above RAkEL method, negative samples that do not belong to any label are usually dominant in a k -label set. The samples could be highly imbalanced. To overcome this difficulty, we apply a combined strategy of over-sampling and under-sampling [59] for data balance.

For a collection of k -label sets L^k , L_i^k is a k -label set belonging to L^k , j is a label representing a subcellular location and $j \in L_i^k$. All the samples that do not locate at any of the localizations appearing in L_i^k are called the negative samples for this set. Usually, the negative sample set S_{neg} is large. If the number of the negative samples $|S_{neg}|$ is twice more than the one from the positive classes S_{max} with the largest number of samples, then the under-sampling is performed. During training, for each positive set S_j representing the samples only at the j -th subcellular location, the over-sampling is made to S_j if $|S_j| < |S_{neg}|$. The dataset balancing strategy is summarized in Algorithm 1.

Algorithm 1**Input:** Initial benchmark dataset D_i for L_i^k **Output:** Balanced benchmark dataset D_i^b for L_i^k **Step 1:** For L_i^k , find the class with the largest number of positive samples $S_{\max}$. Compare

$|S_{\max}|$ with $|S_{\neg}|$. If $|S_{\max}| < \frac{|S_{\neg}|}{2}$, perform under-sampling for $S_{\neg}$. Let $N_{\neg}^b = \left\lceil \left(1 + \frac{|S_{\max}|}{|S_{\neg}|}\right) * |S_{\max}| \right\rceil$ and then randomly select $N_{\neg}^b$ protein sequences from $S_{\neg}$ to compose

a new $S_{\neg}$; Otherwise, the negative set stays unchanged. Let $D_i = D_i \setminus S_{\neg}$.

Step 2: Traverse all labels in L_i^k .**Step 2.1:** For a label $j \in L_i^k$, set $D_i = D_i \setminus S_j$.

Step 2.2: If $|S_j| < \frac{N_{\neg}^b}{2}$, let $N_j^b = \left\lceil \frac{N_{\neg}^b}{|S_j|} * |S_j| \right\rceil$, set S_j^b as an empty set, we use **Step**

2.2.1 to **Step** 2.2.3 to get S_j^b ; Otherwise, $N_j^b = |S_j|$ and select all protein sequences of S_j to form S_j^b .

Step 2.2.1: Let S_{rest} be the dataset of candidate protein sequence, $S_{rest} = S_j$.

Step 2.2.2: If $|S_j^b|$ is equal to N_j^b , then the over-sampling is completed and execute **Step** 2.3; Otherwise, randomly select a sequence s from S_{rest} , set $S_j^b = S_j^b \cup \{s\}$, $S_{rest} = S_{rest} \setminus \{s\}$.

Step 2.2.3: If S_{rest} is empty, to **Step 2.2.1**; Otherwise, to **Step 2.2.2**.

Step 2.3: Sequentially obtain the balanced dataset of all labels in L_i^k , i.e., all subcellular locations.

Step 3: The balanced benchmark dataset $D_i^b = D_i \cup S_{\neg} \cup all\{S_j^b\}$.

2.3 Feature Representation

A protein sequence is represented as:

$$P = R_1 R_2 R_3 \dots R_L \quad (2)$$

where L denotes its length, R_i is the i -th amino residue in the sequence, it has 20 different categories. We first convert symbol sequences into numeric vectors by calculating their statistical properties. Specifically, we integrate Pseudo Amino Acid (PseAA), Amino Acid Index Distribution (AAID), physical and chemical properties (PhyChem) into a fused feature in the paper.

The PseAA consists of two features. The first 20 dimensions represent frequencies of twenty amino acid residues [60]. The rest are m -level correlation factors. It is defined as:

$$y_j = \begin{cases} \frac{f_j}{\sum_{i=1}^{20} f_i + \omega \sum_{n=1}^m \lambda_n} & (1 \leq j \leq 20) \\ \frac{\omega \lambda_{j-20}}{\sum_{i=1}^{20} f_i + \omega \sum_{n=1}^m \lambda_n} & (20+1 \leq j \leq 20+m) \end{cases}, \quad (3)$$

where ω is a weight factor, it is fixed to 0.05 in this paper and f_j is the frequency of amino acid residues. λ_n is the n -level correlation factor defined by:

$$\lambda_n = \frac{1}{L-n} \sum_{i=1}^{L-n} X_{i,i+n} (n < L), \quad (4)$$

where $X_{i,i+n}$ is the correlation between the i -th and $(i+n)$ -th amino residues. The max correlation distance $m < L$.

The AAID features contain three parts. The first 20 dimensions represent composition information, while the rest are second order and third order center distances in the index distribution [61]. They are defined as:

$$d_k = J_i f_i (1 \leq k \leq 20, 1 \leq i \leq 20), \quad (5)$$

$$d_k = \sum_{j=1}^{N_i} \left(\frac{p_{ij} - \bar{p}_i}{L} J_i \right)^2 (21 \leq k \leq 40, 1 \leq i \leq 20), \quad (6)$$

$$d_k = \sum_{j=1}^{N_i} \left(\frac{p_{ij} - \bar{p}_i}{L} J_i \right)^3 (41 \leq k \leq 60, 1 \leq i \leq 20), \quad (7)$$

where J_i is the physical and chemical value of the 20 amino acids, p_{ij} represents the position of the amino acid type r_i in the protein sequence, $\bar{p}_i$ is the average position of r_i , N_i represents the occurrence of r_i in the sequence and L is the length of the protein sequence.

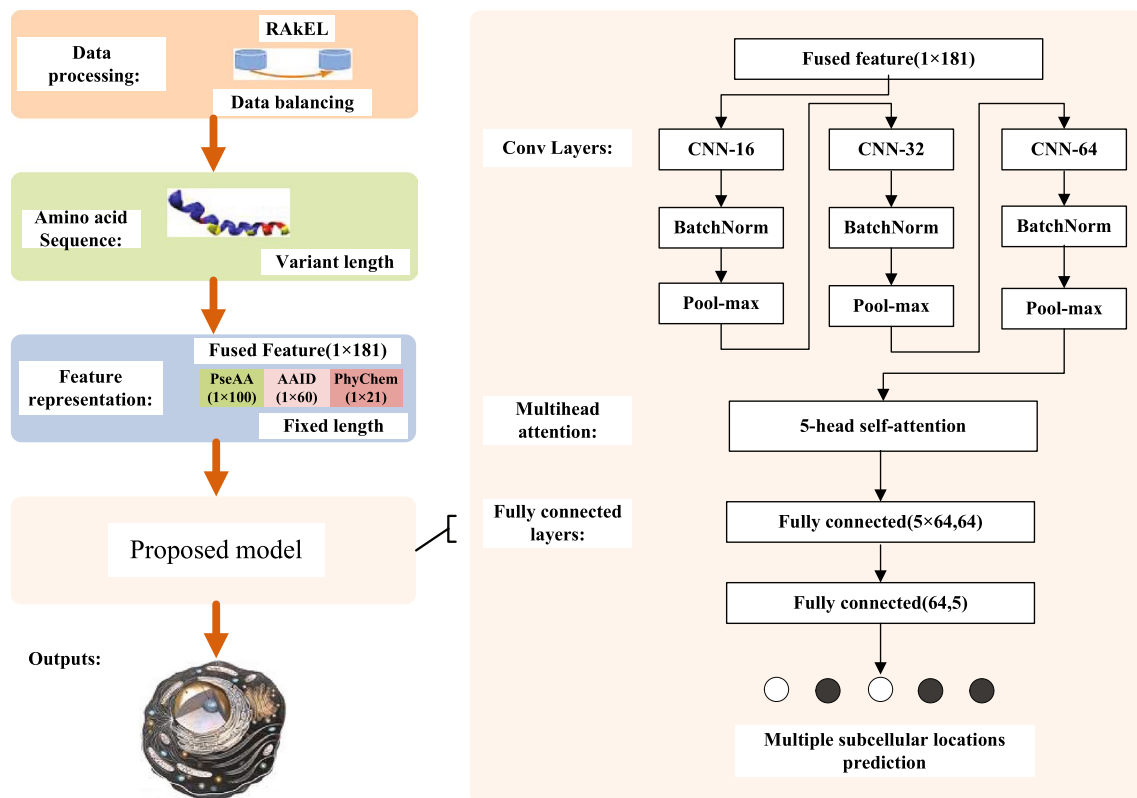


Fig. 2 The flowchart of the proposed method

The PhyChem is based on 7 physical and chemical properties of amino acid residues. Each physical and chemical property is further divided into three categories, i.e., hydrophilic, hydrophobic, and neutral [62]. The definition is given below:

$$s_{i,c} = \frac{N_{i,c}}{L}, \quad (8)$$

where i and c represent the 7 different properties and the 3 categories, respectively. $N_{i,c}$ is the total number of amino acid residues for the i -th property of the c -th category, and L is also the length of the protein sequence.

Besides, we use the min–max normalization method to adjust the different features to the same scale:

$$x^* = \frac{x - \min}{\max - \min}. \quad (9)$$

The fused feature is the concatenation of those features:

$$F = [\text{PseAA}, \text{AAID}, \text{PhyChem}]. \quad (10)$$

2.4 Prediction Model Construction

We use deep convolutional neural network (DCNN) with multi-head self-attention in this paper [63]. The basic convolution module is composed by a convolutional layer [64], a pooling layer [65], and a batch normalization layer. Three basic convolution modules make up DCNN module. Among them, three convolutional layers use different kernels that can match different local patterns [66]. The outputs of convolutional layers pass through the ReLU activation functions to generate a feature map. Then, the feature maps go through a batch normalization layer and a max-pooling layer which keeps primary features [67]. After a stack of convolutional modules [68], high level abstractions can be extracted [69]. The final outputs can be considered as new highly abstracted representations of the original fused feature [70, 71]. The network structure is shown on Fig. 2.

The features produced by the stacked convolution modules are fed into a multi-head self-attention module [72]. Attention mechanism can drive the model to pay more attention to the critical information by assigning different weights during training [73]. Specifically, the multi-head self-attention module receives the output of DCNN module, $Z = [z_1, z_2, \dots, z_M]^T$, where $z_i \in R^D (i = 1, 2, \dots, M)$, D refers to the total number of convolutional filters in last convolutional layer. M is length of the output from DCNN module. To calculate the attention weights, we first need to calculate the energy score matrix E defined as:

$$E = H_2 \times \tan(H_1 Z) \quad (11)$$

where H_1 and H_2 are weight matrices. The shape of H_1 is da -by- D , where da is the attention-dimension. The shape of H_2 is $head$ -by- da , where $head$ is the number of attention heads.

The SoftMax activation function is applied to the energy score matrix E to transform the matrix into a probability distribution as follows:

$$A = \text{softmax}(E), \quad (12)$$

$$a_{ij} = \frac{\exp(e_{ij})}{\sum_{i=1}^M \exp(e_{ij})} (1 \leq i \leq M; 1 \leq j \leq N), \quad (13)$$

where e_{ij} and a_{ij} represent the (i,j) -th element of energy score matrix E and attention weight matrix A , respectively. N is the number of protein sequences.

The j -th column vector of each head is the inner product of the features produced by DCNN and its weights, which is computed as:

$$h_j = \sum_{i=1}^M a_{ij} z_i (1 \leq j \leq N), \quad (14)$$

$$H_h = [h_1, \dots, h_j, \dots, h_N], \quad (15)$$

where H_h is the output feature of a head. In this way, the output features of all heads can be generated. We concatenate all output features together:

$$M = \text{concatenate}(H_h) (1 \leq h \leq \text{head}). \quad (16)$$

After multi-head self-attention, we use a fully connected layer with the sigmoid activation function for final decision making. The output layer with k nodes. k is the number of subcellular locations.

The loss function that we used is a customized cross entropy, defined as follows:

Table 1 The parameters used in experiment

Component	Parameter	value
PseAA	Max level correlation factor	80
	Dimension	100
AAID	Dimension	60
PhyChem	Dimension	21
Fused feature	Dimension	181
Convolutional layers	Kernels	3,5,5
	Filters	16,32,64
Pooling layers	Kernels (1-Dimensional)	2
Self-attention mechanism	heads	5
	Attention-dimension	10
Fully connected layer	Neurons	64

$$\text{Loss} = -\frac{1}{N} \sum_{j=1}^N \sum_{i=1}^k \left\{ \mu \left[y_{ij} \log y_{ij}^{\text{pred}} + (1 - y_{ij}) \log (1 - y_{ij}^{\text{pred}}) \right] \right\}, \quad (17)$$

where N is the number of protein sequences, k is the number of labels, y_{ij} is the i -th true label for j -th sequence, y_{ij}^{pred} is the i -th predicted label for j -th sequence, μ is a coefficient, we use it to balance proportion of positive samples and negative samples in Loss. It can be expressed as:

$$\mu = \begin{cases} 1, y_{ij} = 1 \\ \frac{M_i^+}{M_i^-}, y_{ij} = 0 \end{cases} \quad (18)$$

where M_i^+ and M_i^- are the number of positive samples and negative samples on i -th label, respectively.

2.5 Proposed Method Based on DCNN with Self-Attention Mechanism

The proposed method consists of three major parts. The overall workflow of our model and the corresponding algorithm are listed in Fig. 2 and below, respectively.

Step 1: Dataset processing—RAkEL is used for dataset division. The data balancing strategy is applied to make the training samples distributed evenly across all categories in each k -label set.

Step 2: Feature representation—Three different features are extracted from the input protein sequences and fused to obtain the final integrated features. The fusion feature is the direct input of the DCNN.

Step 3: Prediction model construction—DCNN with a self-attention mechanism is used to extract high-dimensional features and predict the labels of data samples. For each k -label set, a model is trained to perform traditional multi-class classification. Since one subcellular localization

can be in multiple k -label sets, multiple predictions are produced by different classifiers for that localization. The model with best accuracy at that localization is selected as the one in charge of that localization. We get the predicted results of all subcellular locations in the same way while they are the predicted results of the protein sequences.

2.6 Implementation Details

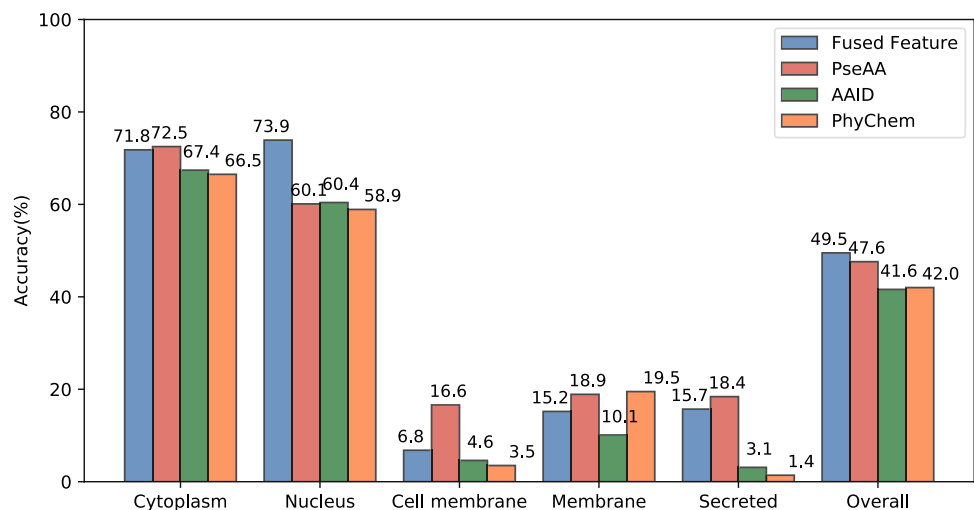
Our deep learning framework is implemented with PyTorch. Model training and testing ran on a workstation with a GTX 1660Ti and 32G RAM. We choose Adam optimizer with a learning rate of 0.001. Meanwhile, a mini-batch strategy is adopted in this paper, which divides data into several small parts by optimizer. The batch size is set to 32. While the max epoch is formed to 20, we stop training when the performance on validation sets begins to decline. A dropout rate of 0.5 is applied in fully connected layers to avoid overfitting. Other parameters are listed in Table 1.

3 Results and Discussion

We design the experiments according to the problems to be solved in this paper (see Sect. 1).

- 1) The experiments in Sect. 3.2 show that the prediction results are highly related to the number of samples using three single features alone or their fused feature, i.e., the classifiers are inclined to the majorities.
- 2) The experiments in Sect. 3.3 show that the critical parameter in RAkEL directly affects the complexity of the prediction task. As the number increases, the correlation between labels becomes more difficult to be dealt.

Fig. 3 Performance of different feature representation methods on unbalanced dataset



- 3) The experiments in Sect. 3.4 show that our pre-processing method using RAKEL and data balancing strategy obviously improves the prediction accuracy.
- 4) The experiments in Sects. 3.4 and 3.5 describe the performance of the proposed method and compare it with other existing subcellular location prediction methods. The results show that our proposed method has effectively solved the data imbalance issue and has better generalization.

3.1 Evaluation Criteria

We employ fivefold Cross Validation [74] to evaluate the performance of classifiers. Specifically, the benchmark dataset is randomly divided into five subsets with equal size, without overlapping. In each turn, we use three subsets as the training set and one subset as the validation set to train the classifiers. The rest one subset which does not appear in the training and validation process is the test set for performance evaluation.

To comprehensively evaluate the performance of the proposed method, we use ACC, F_1 and Recall as the evaluation criteria in our experiments. Their definitions are given below:

$$ACC_p = \frac{TP}{TP + FN}, \quad (19)$$

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}, \quad (20)$$

$$Precision_i = \frac{TP_i}{TP_i + FP_i}, \quad (21)$$

$$Recall_i = \frac{TP_i}{TP_i + FN_i}, \quad (22)$$

$$F_1 = \frac{1}{n} \sum_{i=1}^n \frac{2 \times Precision_i \times Recall_i}{Precision_i + Recall_i}, \quad (23)$$

$$Recall = \frac{1}{n} \sum_{i=1}^n Recall_i, \quad (24)$$

where TP, TN, FP and FN represent true positives, true negatives, false positives, and false negatives, respectively. i represents the i -th subcellular localization. ACC_p is the Accuracy of positive samples while ACC is the accuracy of all samples. F_1 and Recall are measurements for the overall prediction. AUC value from ROC curve is also used for evaluations [75].

3.2 Performance of Different Feature Representation Methods on Unbalanced Dataset

We compared the prediction performance when using three single features alone and their fused feature on an unbalanced dataset, i.e., the dataset has not been processed by RAKEL and data balancing strategy. To demonstrate the impact of data imbalance, we compared the performance on positive samples. The model in this sub-section is DCNN without the attention mechanism. The network structure and the settings of the training parameters are the same as those described in Sect. 2.6. The prediction results are shown in Fig. 3.

As shown in Fig. 3, the model with fused features achieves the best overall accuracy, indicating that it can more

Fig. 4 Performance of different k on RAKEL with data balancing strategy

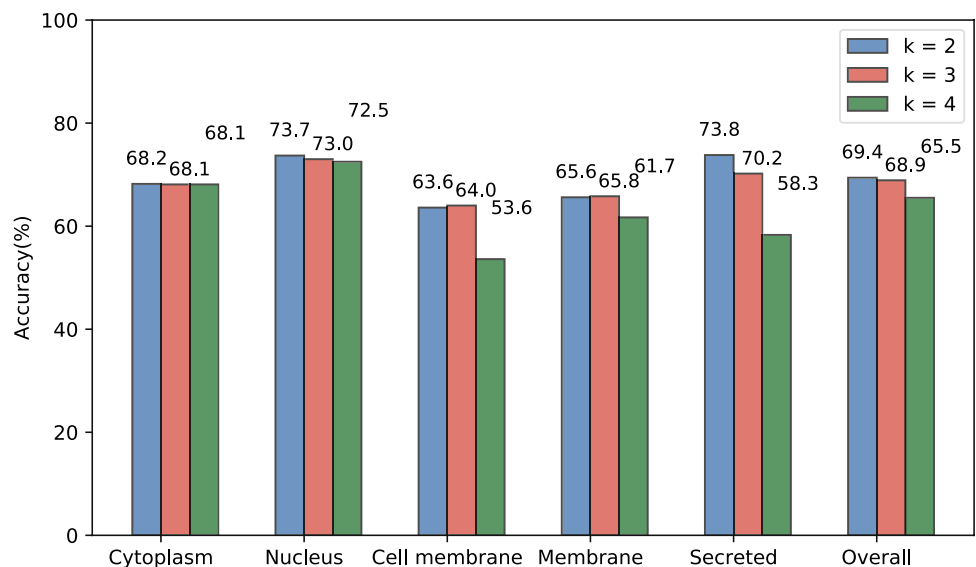
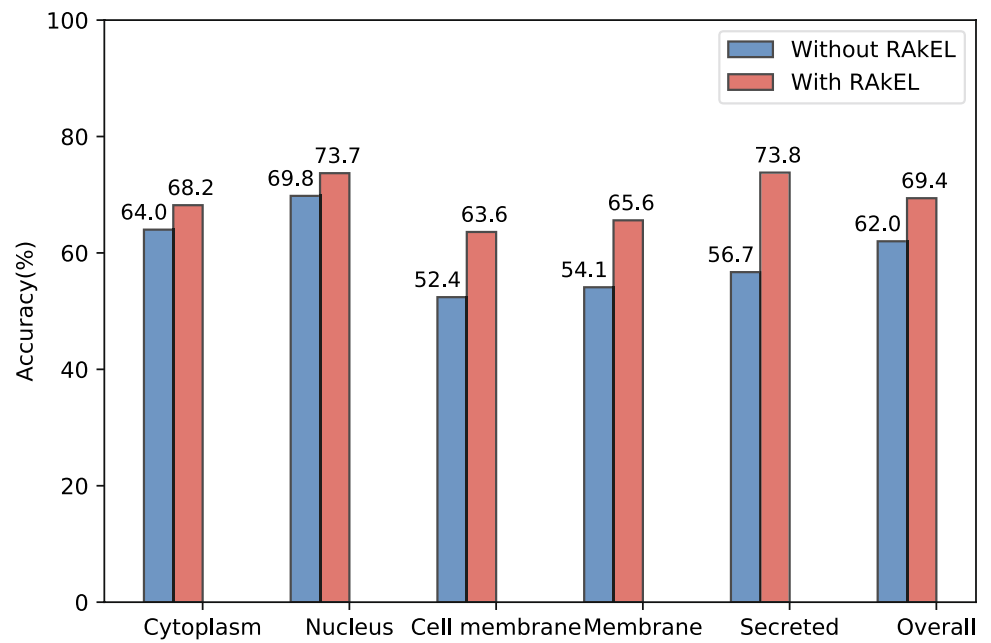
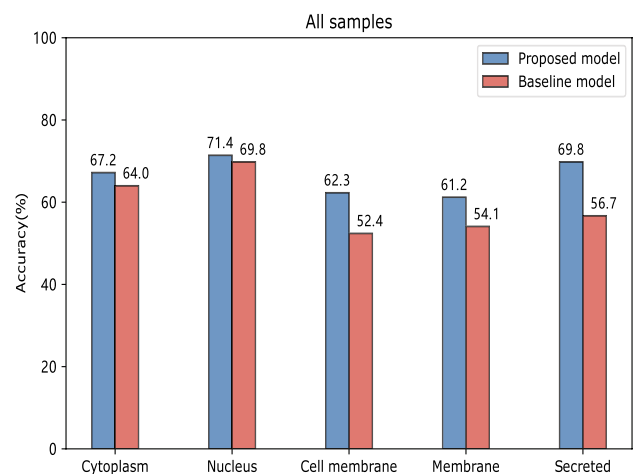
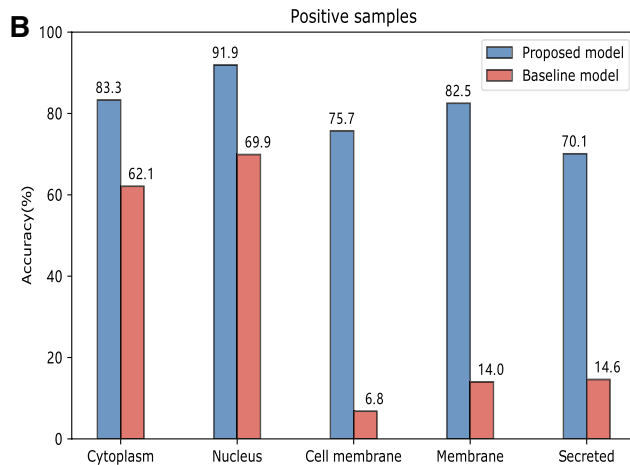
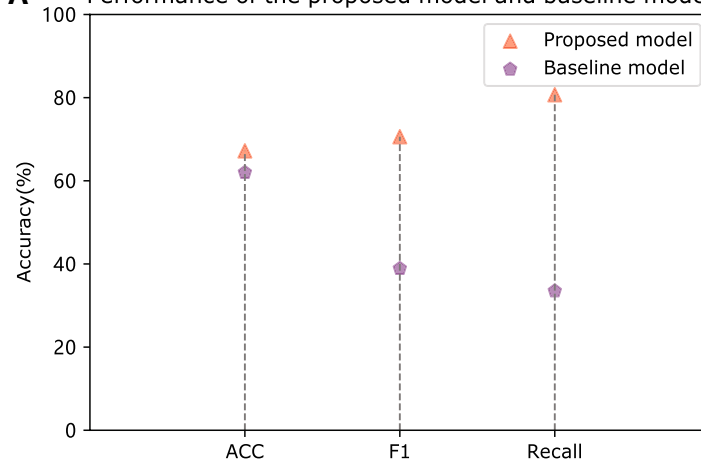


Fig. 5 Performance with and without RAKEL**A** Performance of the proposed model and baseline model**Fig. 6** Performance of the proposed method

comprehensively reflect the characteristics of the amino acid sequences. The ACC_p of all subcellular locations are 49.5%, 47.6%, 41.6% and 42.0% with the fused feature, PseAA, AAID and PhyChem, respectively. The accuracy is very different during each subcellular localization using the same feature representation method, and it is highly related to the number of samples. For instance, the model with the PseAA feature has the highest prediction accuracy value of 72.5% on cytoplasm. Locations with insufficient samples generally have low prediction accuracy, e.g., secreted has the lowest prediction accuracy value of 1.4% when only using the PhyChem feature.

3.3 Comparison Between Different k of Random k -Label SETS

RAkEL can effectively reduce the complexity of classification problems. Therefore, the parameter k in RAkEL is pivotal since it decides the number of prediction labels and the complexity of classification task. In this paper, there are five subcellular locations. The number of classifiers required to cover all subcellular locations increases as k decreases.

We compared the ACC when $k=2, 3$, and 4 on the balanced dataset, respectively. In this experiment, the feature representation method is the fused feature. The structures of DCNNs are consistent, i.e., all of them do not use the self-attention mechanism, and the settings of the structural and training parameters are the same as those described in Sect. 2.6. The results are illustrated in Fig. 4.

As a result, the overall ACC of RAkEL with 2 labels is higher than that of 3 labels or 4 labels. On individual subcellular locations, RAkEL with 2 labels obtains the highest accuracy in 3 locations out of the 5 locations, while RAkEL with 3 labels achieves the best performance in the other 2 locations. The ACC of RAkEL with 4 labels is consistently lower than or equal to the other cases in all 5 locations. It is because the number of labels directly affects the complexity of the prediction task. As the number increases, the correlation between labels increases exponentially, which would be more difficult to be dealt.

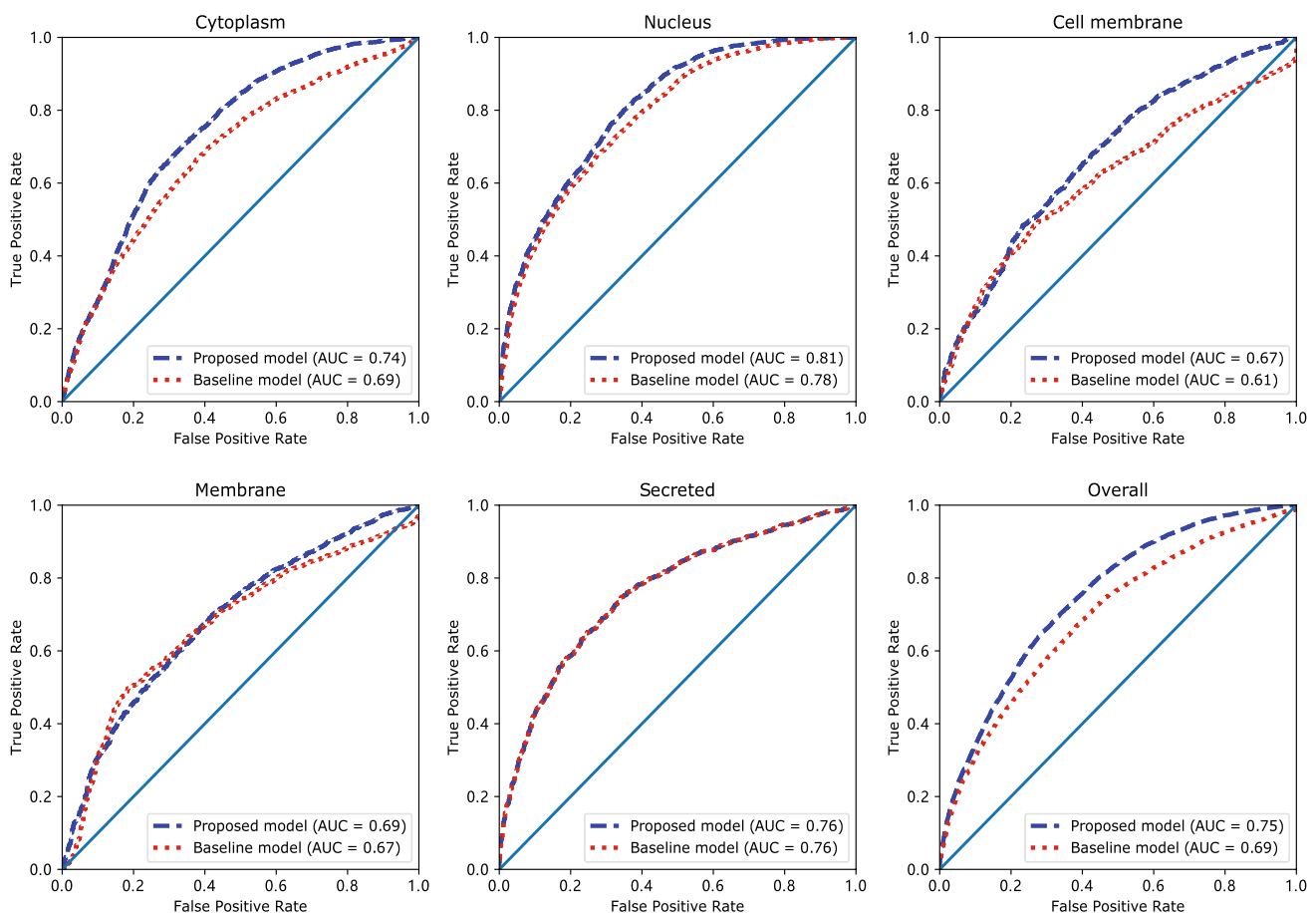


Fig. 7 The ROCs of proposed model and baseline model on the whole dataset as well as each subcellular localization for all samples

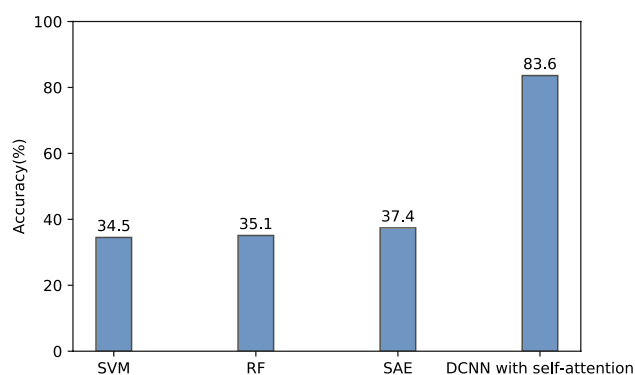


Fig. 8 Performance of different classifiers

3.4 Comparison the Performance With and Without RAKEL

Table 2 Performance of proposed method and SAE on subcellular location

Subcellular location	Proposed method (%)	SAE (%)
Cytoplasm	83.3	45.9
Nucleus	91.9	53.6
Cell membrane	75.7	31.2
Membrane	82.5	30.6
Secreted	70.1	47.8
Overall	83.6	37.4

A comparison between the models with and without RAKEL and the data balancing strategy is conducted, where the prediction accuracy on each subcellular localization is evaluated. The feature representation method is the fused feature. The parameter k in RAKEL is set to 2. The structure and parameter settings of DCNN are the same as those in the previous sub-section. The evaluation criterion is ACC. The prediction results are shown in Fig. 5.

With RAKEL, the overall ACC of prediction is up to 69.4%, while without it, the overall ACC lowers to 62.0%. A similar situation also appears at the single subcellular locations. It is more obvious for minority subcellular localizations. For instance, the ACC of membrane localization drops from 65.6% to 54.1% when RAKEL has been removed. For secreted localization, the ACC drops dramatically from 73.8 to 56.7%.

3.5 Performance of the Proposed Method

In this section, we investigated whether the proposed method could promote prediction performance. The baseline model is a DCNN with the same structure as in the proposed model. However, this model does not contain the self-attention layer, and the data is not preprocessed by RAKEL. The structure and parameter settings are as same as those defined in the previous experiment. Besides, they also used the fused feature as inputs. We illustrated the ACC, $F1$, and Recall of both the proposed model and the baseline model in Fig. 6A. The ACC, $F1$ and Recall are all improved by our proposed method, where they are promoted from 62.0% to 67.2%,

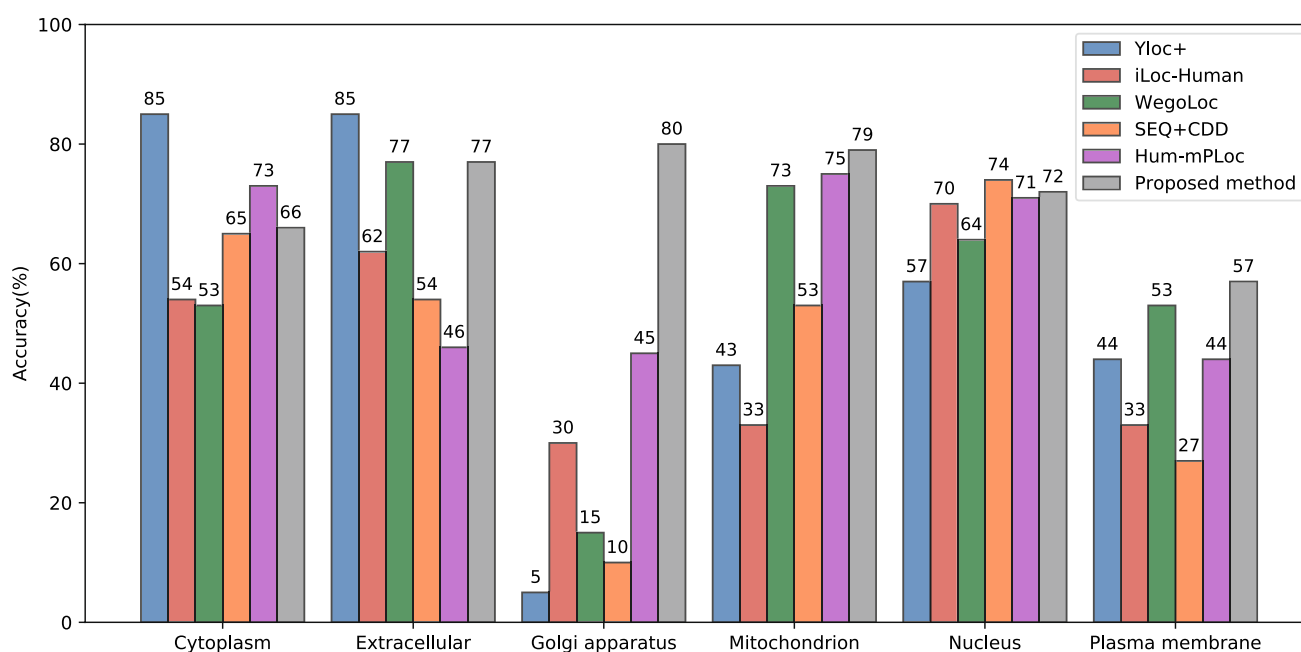


Fig. 9 Performance of different prediction methods

Table 3 Comparison of six prediction methods

Method	ACC	F1
Yloc +	0.45	0.34
iLoc-Human	0.41	0.32
WegoLoc	0.5	0.44
SEQ + CDD	0.47	0.29
Hum-mPLoc	0.63	0.65
Proposed method	0.61	0.41

38.9% to 70.6%, 33.5% to 80.7%, respectively. Notably, the significant improvement on *F1* and Recall indicates that our method can enhance the prediction performance by taking characteristics of the positive samples during classification.

Figure 6B illustrates the comparison results of accuracy obtained by the two models on the five subcellular locations. The left is prediction accuracy for positive samples, and the right is for all samples. As we can see, the prediction performances on all the subcellular localization are improved by the proposed method both for positive and all samples. Especially, our proposed method considerably improves the localization accuracy for positive samples at minority subcellular locations, e.g., the ACC of the normal method for positive samples on secreted is only 14.6%, whereas the proposed method on this location is 70.1%.

In addition, we also exhibited the ROC curves for these two models on each subcellular location for all samples in Fig. 7. The accuracy and AUC values indicate that the proposed method can improve the prediction accuracy of positive samples while also ensuring the accuracy of all samples.

3.6 Comparison with Other Methods

The prediction performance of the proposed method with other classifiers is compared using the same benchmark dataset. There are three representative classifiers, i.e., stacked auto-encoder (SAE), support vector machine (SVM) and random forest classifier (RF) [45] for comparison. The SAE is a hierarchical network that comprises multiple auto-encoder layers, where each auto-encoder has three layers. The above models are constructed and trained according to the referenced literature, except for the dimension adjustment for the input and output. We compared the accuracy on each subcellular location of positive samples. The results are demonstrated in Fig. 8.

Our proposed method achieves the highest ACC_p of 83.6% among the four classifiers for all subcellular locations in the benchmark dataset, while SVM, RF and SAE only obtain the accuracy of 34.5%, 35.1%, and 37.4%, respectively. Since SAE slightly outperformed SVM and RF, we further compared it with our method to each subcellular location, as shown in Table 2. Obviously, the proposed method is significantly better than SAE, especially on

minority categories. For instance, SAE has a poor prediction result on cell membrane and membrane. They are only 31.2% and 30.6%, respectively. In contrast, the accuracy of our method on these two subcellular locations is 75.7% and 82.5%, respectively.

And then, we compared our proposed method with five existing subcellular location prediction methods specialized for human proteins, including YLOC + [76], iLoc-Hum [77], WegoLoc [78], Hum-mPLoc 3.0 and SEQ + CDD [15]. The first four methods use GO features, while the last one uses sequence and CDD features. Since the HumB dataset and HumT dataset contain the subcellular locations involved in the above methods, we also used them for training and testing for our proposed method. During the experiments, the sequences in HumT dataset are submitted the online servers of YLOC +, iLoc-Hum, WegoLoc and SEQ + CDD to get predictions. The HumB dataset is divided into a training set (80%) and a validation set (20%) to train Hum-mPLoc 3.0 and our method. The HumT dataset is used as an extra independent testing set to evaluate the model generalization. We compared each subcellular location of positive samples and two evaluation criteria of all samples. The results are demonstrated in Fig. 9 and Table 3, respectively.

As shown in Fig. 9, the proposed method obtains better prediction accuracy for positive samples. It achieves the best accuracy at three subcellular localizations, i.e., Golgi apparatus, mitochondrion, and plasma membrane. For these subcellular localizations, the ACC_p are 80%, 77%, and 57%, respectively. Especially on Golgi apparatus, its prediction accuracy for positive samples far exceeds other methods. Its prediction accuracy at the other three subcellular localizations is competitive, without any considerable degradation. Notably, the prediction accuracy of our proposed method at each subcellular location is relatively stable. In contrast, other methods are greatly influenced by the number of samples, e.g., YLoc + achieves the highest prediction accuracy on cytoplasm and extracellular, but also gets the worst prediction result on Golgi apparatus.

In Table 3, it can be seen that the proposed method surpasses the other four algorithms except Hum-mPLoc. The reason may be that Hum-mPLoc uses the latest version of annotation databases, which contains relatively complete GO features for the HumT dataset. Although YLOC +, iLoc-Hum and WegoLoc also employ GO features, the reason why they did not obtain satisfactory performances might be that they use an earlier version of the annotation databases in the training process. In comparison, the proposed method achieves much more steady and higher ACC and *F1* on all subcellular localizations, indicating that it has effectively solved the data imbalance issue and has better generalization.

3.7 Discussion

Accurate prediction of multiple protein subcellular locations is more challenging than the single location prediction due to the complex relationship among different categories. Therefore, the existing feature representation methods cannot fully express the complex relationship between them. The feature fusion method would be better for capturing the relevance between different categories. However, it might also lead to feature redundancy. This paper uses DCNN with a self-attention mechanism to extract high-dimensional features from the fused feature, removing the redundancy of preliminary features, better representing the raw sequences. The DCNN can extract high-dimensional features, and it amplifies the discriminative features using the convolutional layer. The self-attention mechanism further helps extract critical features. The experimental results demonstrate that the proposed method can improve prediction accuracy.

In addition, we use RAKEL to reduce the complexity of the localization problem by decreasing prediction labels every time. The main parameter of RAKEL is the number of labels, and we compare the localization results with the different number of labels. As a result, better localization results can be obtained when the value reduces. In the experiments, we find that using the RAKEL may increase the number of negative samples in the group, exacerbating the imbalance in multiple protein subcellular localization. To overcome this issue, we apply a combined strategy of over-sampling and under-sampling to ensure data balancing. When the number of negative samples is more than a threshold value, under-sampling is used to them. When the number of positive samples is less than a threshold value, over-sampling is used to them. The results show that this could be a promising strategy, although much work is still needed to research on selecting the threshold value.

We use data processing and DCNN with a self-attention mechanism to improve the prediction accuracy for multiple protein subcellular locations. Although the localization accuracy has been significantly improved, the proposed method is still not well explained biologically. At present, it is generally considered from the perspective of data distribution. In future work, we will attempt to develop methods with biological insights. Besides, we use multiple convolutional layers to extract the high-dimensional features. We expect to collect more labeled samples and expand the benchmark dataset to study the structure of DCNN further.

4 Conclusion

In this paper, we present an improved model for multiple protein subcellular locations prediction based on DCNN with a self-attention mechanism. We use RAKEL to reduce

the number of prediction labels. And we simultaneously apply a combined strategy of over-sampling and under-sampling to ensure balancing. Besides, we use the feature fusion method to extract preliminary features of protein sequences. Then, high-level abstract features are extracted through DCNN with the self-attention mechanism to remove the redundancy in the fused feature. Our results have demonstrated the efficacy of the proposed method. Future challenges include exploring the model from the biological perspective and designing more adaptive network architectures.

Acknowledgements This research was supported by the National Natural Science Foundation of China (61876102), Shandong Provincial Natural Science Foundation (ZR2021MF036) and University Innovation Team Project of Jinan (2019GXRC015).

Declarations

Conflict of interest The authors declare that there is no conflict of interest regarding the publication of this paper.

References

1. Jha BK, Joshi H, Dave DD (2018) Portraying the effect of calcium-binding proteins on cytosolic calcium concentration distribution fractionally in nerve cells. *Interdiscip Sci Comput Life Sci* 10(4):674–685. <https://doi.org/10.1007/s12539-016-0202-7>
2. Chen X, Shi SP, Suo SB, Xu HD, Qiu JD (2015) Proteomic analysis and prediction of human phosphorylation sites in subcellular level reveal subcellular specificity. *Bioinformatics* 31(2):194–200. <https://doi.org/10.1093/bioinformatics/btu598>
3. Zhang W, Xue X, Xie C, Li Y, Liu J, Chen H, Li G (2021) CEGSO: boosting essential proteins prediction by integrating protein complex, gene expression, gene ontology, subcellular localization and orthology information. *Interdiscip Sci Comput Life Sci*. <https://doi.org/10.1007/s12539-021-00426-7>
4. Thul PJ, Åkesson L, Wiking M, Mahdessian D, Geladaki A, Blal HA et al (2017) A subcellular map of the human proteome. *Science*. <https://doi.org/10.1126/science.aal3321>
5. Yao L, Wang H, Bin Y (2021) Predicting hot spot residues at protein–DNA binding interfaces based on sequence information. *Interdiscip Sci Comput Life Sci* 13(1):1–11. <https://doi.org/10.1007/s12539-020-00399-z>
6. Ludwik KA, von Kuegelgen N, Chekulaeva M (2019) Genome-wide analysis of RNA and protein localization and local translation in mESC-derived neurons. *Methods* 162:31–41. <https://doi.org/10.1016/j.ymeth.2019.02.002>
7. Anusha AR, Vinodchandra SS (2017) Probabilistic neural network inferences on oligonucleotide classification based on oligo: target interaction. *Asian conference on intelligent information and database systems*. Springer, Cham, pp 733–740. https://doi.org/10.1007/978-3-319-54430-4_70
8. Xue L, Tang XQ (2021) A new framework for discovering protein complex and disease association via mining multiple databases. *Interdiscip Sci Comput Life Sci*. <https://doi.org/10.1007/s12539-021-00432-9>
9. Kulmanov M, Hoehndorf R (2020) DeepGOPlus: improved protein function prediction from sequence. *Bioinformatics* 36(2):422–429. <https://doi.org/10.1093/bioinformatics/btz595>

10. Mazmanian K, Sargsyan K, Lim C (2020) How the local environment of functional sites regulates protein function. *J Am Chem Soc* 142(22):9861–9871. <https://doi.org/10.1021/jacs.0c02430>
11. Corrêa T, Feltes BC, Gonzalez EA, Baldo G, Matte U (2021) Network analysis reveals proteins associated with aortic dilatation in *Mucopolysaccharidoses*. *Interdiscip Sci Comput Life Sci* 13(1):34–43. <https://doi.org/10.1007/s12539-020-00406-3>
12. Zhang S (2018) Prediction of protein subcellular localization with oversampling approach and Chou's general PseAAC. *J Theor Biol* 437:239–250. <https://doi.org/10.1016/j.jtbi.2017.10.030>
13. Yang Y, Wu W, Liu T, Tong L, Lijun D, Hetian L (2021) A robust method for protein depletion based on gene editing. *Methods* 194:3–11. <https://doi.org/10.1016/j.ymeth.2021.03.001>
14. Aswathy AL, Hareendran A (2021) COVID-19 diagnosis and severity detection from CT-images using transfer learning and back propagation neural network. *J Infect Public Health* 14(10):1435–1445. <https://doi.org/10.1016/j.jiph.2021.07.015>
15. Zhou H, Yang Y, Shen HB (2017) Hum-mPLoc 3.0: prediction enhancement of human protein subcellular localization through modeling the hidden correlations of gene ontology and functional domain features. *Bioinformatics* 33(6):843–853. <https://doi.org/10.1093/bioinformatics/btw723>
16. Singh S, Singh DB, Singh A, Gautam B, Ram G, Dwivedi S, Rameke PW (2016) An approach for identification of novel drug targets in *Streptococcus pyogenes* SF370 through pathway analysis. *Interdiscip Sci Comput Life Sci* 8(4):388–394. <https://doi.org/10.1007/s12539-015-0139-2>
17. Wan S, Mak MW, Kung SY (2016) Sparse regressions for predicting and interpreting subcellular localization of multi-label proteins. *BMC Bioinf* 17(1):1–17. <https://doi.org/10.1186/s12859-016-0940-x>
18. Xiang Q, Liao B, Li X, Xu H, Chen J, Shi Z et al (2017) Subcellular localization prediction of apoptosis proteins based on evolutionary information and support vector machine. *Artif Intell Med* 78:41–46. <https://doi.org/10.1016/j.artmed.2017.05.007>
19. Shen Y, Tang J, Guo F (2019) Identification of protein subcellular localization via integrating evolutionary and physicochemical information into Chou's general PseAAC. *J Theor Biol* 462:230–239. <https://doi.org/10.1016/j.jtbi.2018.11.012>
20. Cheng X, Xiao X, Chou KC (2018) pLoc-mEuk: Predict subcellular localization of multi-label eukaryotic proteins by extracting the key GO information into general PseAAC. *Genomics* 110(1):50–58. <https://doi.org/10.1016/j.ygeno.2017.08.005>
21. Du L, Meng Q, Chen Y, Wu P (2020) Subcellular location prediction of apoptosis proteins using two novel feature extraction methods based on evolutionary information and LDA. *BMC Bioinf* 21:1–19. <https://doi.org/10.1186/s12859-020-3539-1>
22. Wan S, Mak MW (2015) Machine learning for protein subcellular localization prediction. *De Gruyter*
23. Behbahani M, Nosrati M, Moradi M, Mohabatkar H (2020) Using Chou's general pseudo amino acid composition to classify lacases from bacterial and fungal sources via Chou's five-step rule. *Appl Biochem Biotechnol* 190(3):1035–1048. <https://doi.org/10.1007/s12010-019-03141-8>
24. Freeman BT, Sokolowski M, Roy-Engel AM, Smither ME, Belancio VP (2019) Identification of charged amino acids required for nuclear localization of human L1 ORF1 protein. *Mob DNA* 10(1):20. <https://doi.org/10.1186/s13100-019-0159-2>
25. Yang J, Zhang Y (2015) I-TASSER server: new development for protein structure and function predictions. *Nucleic Acids Res* 43(W1):W174–W181. <https://doi.org/10.1093/nar/gkv342>
26. Tessier TM, MacNeil KM, Mymryk JS (2020) Piggybacking on classical import and other non-classical mechanisms of nuclear import appear highly prevalent within the human proteome. *Biology* 9(8):188. <https://doi.org/10.3390/biology9080188>
27. Tahir M, Khan A, Kaya H (2014) Protein subcellular localization in human and hamster cell lines: employing local ternary patterns of fluorescence microscopy images. *J Theor Biol* 340:85–95. <https://doi.org/10.1016/j.jtbi.2013.08.017>
28. Kumar R, Kumari B, Kumar M (2018) Proteome-wide prediction and annotation of mitochondrial and sub-mitochondrial proteins by incorporating domain information. *Mitochondrion* 42:11–22. <https://doi.org/10.1016/j.mito.2017.10.004>
29. Yadav AK, Singla D (2020) VacPred: sequence-based prediction of plant vacuole proteins using machine-learning techniques. *J Biosci* 45(1):1–9. <https://doi.org/10.1007/s12038-020-00076-9>
30. Zhao L, Wang J, Nabil MM, Zhang J (2018) Deep Forest-based prediction of protein subcellular localization. *Curr Gene Ther* 18(5):268–274. <https://doi.org/10.2174/1566523218666180913110949>
31. Almagro Armenteros JJ, Sønderby CK, Sønderby SK, Nielsen H, Winther O (2017) DeepLoc: prediction of protein subcellular localization using deep learning. *Bioinformatics* 33(21):3387–3395. <https://doi.org/10.1093/bioinformatics/btx431>
32. Long W, Yang Y, Shen HB (2020) ImPLoc: a multi-instance deep learning model for the prediction of protein subcellular localization based on immunohistochemistry images. *Bioinformatics* 36(7):2244–2250. <https://doi.org/10.1093/bioinformatics/btz909>
33. Ashkenazy H, Abadi S, Martz E, Chay O, Mayrose I, Pupko T, Ben-Tal N (2016) ConSurf 2016: an improved methodology to estimate and visualize evolutionary conservation in macromolecules. *Nucleic Acids Res* 44(W1):W344–W350. <https://doi.org/10.1093/nar/gkw408>
34. Suma LS, Vinod Chandra SS (2021) Mining of structural motifs in proteins using artificial bee colony optimization framework for drugability. *J Bioinform Comput Biol*. <https://doi.org/10.1142/S0219720021500256>
35. Mandal M, Mukhopadhyay A, Maulik U (2015) Prediction of protein subcellular localization by incorporating multiobjective PSO-based feature subset selection into the general form of Chou's PseAAC. *Med Biol Eng Comput* 53(4):331–344. <https://doi.org/10.1007/s11517-014-1238-7>
36. Younis H, Anwar MW, Khan MUG, Sikandar A, Bajwa UI (2021) A new sequential forward feature selection (SFFS) algorithm for mining best topological and biological features to predict protein complexes from protein-protein interaction networks (PPINs). *Interdiscip Sci Comput Life Sci*. <https://doi.org/10.1007/s12539-021-00433-8>
37. Chandra V, Girijadevi R, Nair AS, Pillai SS, Pillai RM (2010) MTar: a computational microRNA target prediction architecture for human transcriptome. *BMC Bioinf* 11(1):1–9. <https://doi.org/10.1186/1471-2105-11-S1-S2>
38. Wei L, Xing P, Shi G, Ji Z, Zou Q (2017) Fast prediction of protein methylation sites using a sequence-based feature selection technique. *IEEE/ACM Trans Comput Biol Bioinf* 16(4):1264–1273. <https://doi.org/10.1109/TCBB.2017.2670558>
39. Arango-Argoty GA, Jaramillo-Garzón JA, Castellanos-Domínguez G (2015) Feature extraction by statistical contact potentials and wavelet transform for predicting subcellular localizations in gram negative bacterial proteins. *J Theor Biol* 364:121–130. <https://doi.org/10.1016/j.jtbi.2014.08.051>
40. Javed F, Hayat M (2019) Predicting subcellular localization of multi-label proteins by incorporating the sequence features into Chou's PseAAC. *Genomics* 111(6):1325–1332. <https://doi.org/10.1016/j.ygeno.2018.09.004>
41. Qu X, Wang D, Chen Y, Qiao S, Zhao Q (2015) Predicting the subcellular localization of proteins with multiple sites based on multiple features fusion. *IEEE/ACM Trans Comput Biol Bioinf* 13(1):36–42. <https://doi.org/10.1109/TCBB.2015.2485207>
42. Cheng X, Xiao X, Chou KC (2017) pLoc-mPlant: predict subcellular localization of multi-location plant proteins by incorporating

- the optimal GO information into general PseAAC. *Mol BioSyst* 13(9):1722–1727. <https://doi.org/10.1039/C7MB00267J>
43. Liu Z, Hu J (2016) Mislocalization-related disease gene discovery using gene expression based computational protein localization prediction. *Methods* 93:119–127. <https://doi.org/10.1016/j.ymeth.2015.09.022>
 44. Chou KC (2019) Advances in predicting subcellular localization of multi-label proteins and its implication for developing multi-target drugs. *Curr Med Chem* 26(26):4918–4943. <https://doi.org/10.2174/0929867326666190507082559>
 45. Wei L, Ding Y, Su R, Tang J, Zou Q (2018) Prediction of human protein subcellular localization using deep learning. *J Parallel Distrib Comput* 117:212–217. <https://doi.org/10.1016/j.jpdc.2017.08.009>
 46. Cheng X, Xiao X, Chou KC (2018) pLoc-mHum: predict subcellular localization of multi-location human proteins via general PseAAC to winnow out the crucial GO information. *Bioinformatics* 34(9):1448–1456. <https://doi.org/10.1093/bioinformatics/btx711>
 47. Rawat W, Wang Z (2017) Deep convolutional neural networks for image classification: a comprehensive review. *Neural Comput* 29(9):2352–2449. https://doi.org/10.1162/neco_a_00990
 48. Sui X, Zheng Y, Wei B, Bi H, Wu J, Pan X et al (2017) Choroid segmentation from optical coherence tomography with graph-edge weights learned from deep convolutional neural networks. *Neurocomputing* 237:332–341. <https://doi.org/10.1016/j.neucom.2017.01.023>
 49. Lee CY, Gallagher PW, Tu Z (2016) Generalizing pooling functions in convolutional neural networks: mixed, gated, and tree. In: *Artificial intelligence and statistics*. PMLR, pp. 464–472
 50. Tsoumakas G, Katakis I, Vlahavas I (2010) Random k-labelsets for multilabel classification. *IEEE Trans Knowl Data Eng* 22(7):1079–1089. <https://doi.org/10.1109/TKDE.2010.164>
 51. Yapp EK, Li X, Lu WF, Tan PS (2020) Comparison of base classifiers for multi-label learning. *Neurocomputing* 394:51–60. <https://doi.org/10.1016/j.neucom.2020.01.102>
 52. Thabtah F, Hammoud S, Kamalov F, Gonsalves A (2020) Data imbalance in classification: experimental evaluation. *Inf Sci* 513:429–441. <https://doi.org/10.1016/j.ins.2019.11.004>
 53. Zhao B, Zhang X, Li H, Yang Z (2020) Intelligent fault diagnosis of rolling bearings based on normalized CNN considering data imbalance and variable working conditions. *Knowl-Based Syst* 199:105971. <https://doi.org/10.1016/j.knosys.2020.105971>
 54. Wang D, Zhang Z, Jiang Y, Mao Z, Wang D, Lin H, Dong XuD (2021) DM3Loc: multi-label mRNA subcellular localization prediction and analysis based on multi-head self-attention mechanism. *Nucleic Acids Res* 49(8):46. <https://doi.org/10.1093/nar/gkab016>
 55. Boutet E, Lieberherr D, Tognolli M, Schneider M, Bairoch A (2007) Uniprotkb/swiss-prot. In: *Plant bioinformatics*. Humana Press, pp. 89–112. https://doi.org/10.1007/978-1-59745-535-0_4
 56. Fu L, Niu B, Zhu Z, Wu S, Li W (2012) CD-HIT: accelerated for clustering the next-generation sequencing data. *Bioinformatics* 28(23):3150–3152. <https://doi.org/10.1093/bioinformatics/bts565>
 57. Huang C, Yuan J (2013) Using radial basis function on the general form of Chou's pseudo amino acid composition and PSSM to predict subcellular locations of proteins with both single and multiple sites. *Biosystems* 113(1):50–57. <https://doi.org/10.1016/j.biosystems.2013.04.005>
 58. Wu YP, Lin HT (2017) Progressive random k-labelsets for cost-sensitive multi-label classification. *Mach Learn* 106(5):671–694. <https://doi.org/10.1007/s10994-016-5600-x>
 59. Ghosh SK, Ghosh A (2020) A novel human diabetes biomarker recognition approach using fuzzy rough multigranulation nearest neighbour classifier model. *Interdiscip Sci Comput Life Sci* 12(4):461–475. <https://doi.org/10.1007/s12539-020-00391-7>
 60. Chou KC (2001) Prediction of protein cellular attributes using pseudo-amino acid composition. *Proteins Struct Funct Bioinform* 43(3):246–255. <https://doi.org/10.1002/prot.1035>
 61. Li B, Cai L, Liao B, Fu X, Bing P, Yang J (2019) Prediction of protein subcellular localization based on fusion of multi-view features. *Molecules* 24(5):919. <https://doi.org/10.3390/molecules24050919>
 62. Pal MK, Lahiri T, Kumar R (2020) ProtPCV: a fixed dimensional numerical representation of protein sequence to significantly reduce sequence search time. *Interdiscip Sci Comput Life Sci* 12(3):276–287. <https://doi.org/10.1007/s12539-020-00380-w>
 63. Mohapatra S, Nayak J, Mishra M, Pati GK, Naik B, Swarnkar T (2021) Wavelet transform and deep convolutional neural network-based smart healthcare system for gastrointestinal disease detection. *Interdiscip Sci Comput Life Sci*. <https://doi.org/10.1007/s12539-021-00417-8>
 64. Long H, Wang M, Fu H (2017) Deep convolutional neural networks for predicting hydroxyproline in proteins. *Curr Bioinform* 12(3):233–238. <https://doi.org/10.2174/1574893612666170221152848>
 65. Al-Ajlan A, El Allali A (2019) CNN-MGP: Convolutional neural networks for metagenomics gene prediction. *Interdiscip Sci Comput Life Sci* 11(4):628–635. <https://doi.org/10.1007/s12539-018-0313-4>
 66. Wang S, Chen H (2019) A novel deep learning method for the classification of power quality disturbances using deep convolutional neural network. *Appl Energy* 235:1126–1140. <https://doi.org/10.1016/j.apenergy.2018.09.160>
 67. Huang W, Cheng J, Yang Y, Guo G (2019) An improved deep convolutional neural network with multi-scale information for bearing fault diagnosis. *Neurocomputing* 359:77–92. <https://doi.org/10.1016/j.neucom.2019.05.052>
 68. Kamel A, Sheng B, Yang P, Li P, Shen R, Feng DD (2018) Deep convolutional neural networks for human action recognition using depth maps and postures. *IEEE Trans Syst Man Cybernet Syst* 49(9):1806–1819. <https://doi.org/10.1109/TSMC.2018.2850149>
 69. Xia KJ, Yin HS, Wang JQ (2019) A novel improved deep convolutional neural network model for medical image fusion. *Clust Comput* 22(1):1515–1527. <https://doi.org/10.1007/s10586-018-2026-1>
 70. Sellami A, Hwang H (2019) A robust deep convolutional neural network with batch-weighted loss for heartbeat classification. *Expert Syst Appl* 122:75–84. <https://doi.org/10.1016/j.eswa.2018.12.037>
 71. Zhou DX (2020) Theory of deep convolutional neural networks: Downsampling. *Neural Netw* 124:319–327. <https://doi.org/10.1016/j.neunet.2020.01.018>
 72. Liu J, Gong X (2019) Attention mechanism enhanced LSTM with residual architecture and its application for protein–protein interaction residue pairs prediction. *BMC Bioinform* 20(1):1–11. <https://doi.org/10.1186/s12859-019-3199-1>
 73. Lin Z, Feng M, Santos CND, Yu M, Xiang B, Zhou B, Bengio Y (2017) A structured self-attentive sentence embedding. *arXiv preprint*. <https://arxiv.org/abs/1703.03130v1>
 74. Roberts DR, Bahn V, Ciuti S, Boyce MS, Elith J, Guillera-Arroita G et al (2017) Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* 40(8):913–929. <https://doi.org/10.1111/ecog.02881>
 75. Kang Q, Meng J, Shi W, Luan Y (2021) Ensemble deep learning based on multi-level information enhancement and greedy fuzzy decision for plant miRNA–lncRNA interaction prediction. *Interdiscip Sci Comput Life Sci*. <https://doi.org/10.1007/s12539-021-00434-7>
 76. Briesemeister S, Rahnenf J, Kohlbacher O (2010) YLoc—an interpretable web server for predicting subcellular localization. *Nucleic Acids Res* 38(2):497–502. <https://doi.org/10.1093/nar/gkq477>

77. Chou KC, Wu ZC, Xiao X (2012) iLoc-Hum: using the accumulation-label scale to predict subcellular locations of human proteins with both single and multiple sites. *Mol BioSyst* 8(2):629–641. <https://doi.org/10.1039/C1MB05420A>
78. Chi SM, Nam D (2012) WegoLoc: accurate prediction of protein subcellular localization using weighted Gene Ontology terms. *Bioinformatics* 28(7):1028–1030. <https://doi.org/10.1093/bioinformatics/bts062>